

NANYANG JUNIOR COLLEGE
JC 2 Preliminary Examination
Higher 2

CANDIDATE
NAME

CLASS

BIOLOGY

9744/02

Paper 2 Structured Questions

13 September 2022

Candidates answer on the Question Paper.

No Additional Materials are required.

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name and CT on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions in the spaces provided on the Question Paper

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
4	
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6	
7	
8	
9	
10	
Total	

This document consists of **25** printed pages.

[Turn over

Answer **all** the questions.

- 1 The cell surface membrane has a fluid mosaic structure.

(a) Describe what is meant by the term *fluid mosaic*.

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[2]

- (b) In 1934, the biologists Davson and Danielli published their suggestion for the structure of the cell surface membrane, as shown in Fig. 1.1.

They suggested that the membrane was a phospholipid bilayer with a layer of hydrophilic protein on both surfaces.

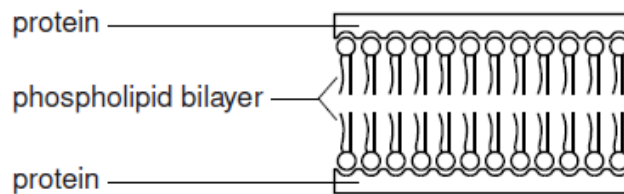


Fig. 1.1

State **one** way in which the Davson-Danielli structure is similar to the fluid mosaic structure **and one** way in which it differs from the fluid mosaic model.

similarity

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difference

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[2]

Fig. 1.2 is a diagram of the structure of a protein channel for ions in a cell surface membrane. Fig. 1.2a shows the channel when open and Fig. 1.2b shows the same channel when closed.

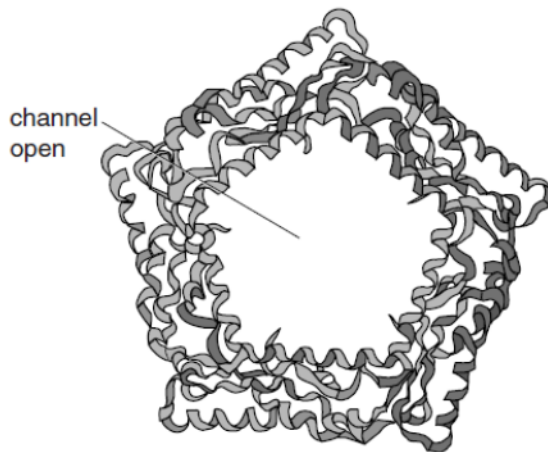


Fig. 1.2a



Fig. 1.2b

(c) (i) Explain why a channel protein is needed for ions to pass across a cell membrane.

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[2]

(ii) Channel proteins are examples of transmembrane proteins. The polypeptides are held together and also interact with phospholipids in the membrane.

Suggest how the polypeptides are held together and how they interact with phospholipids in the membrane.

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[3]

[Total: 9]

- 2 (a) Table 2.1 shows some features of four biological molecules that are all polymers. Complete Table 2.1 by using a tick (✓) to indicate the features that apply to each polymer.

Table 2.1

feature	amylopectin	RNA	polypeptide	phospholipid
synthesised from amino acid monomers				
polymer is branched				
contains nitrogen				
can be found in both animals and plants				

[4]

Fig 2.1 is a diagram of a glycogen molecule.

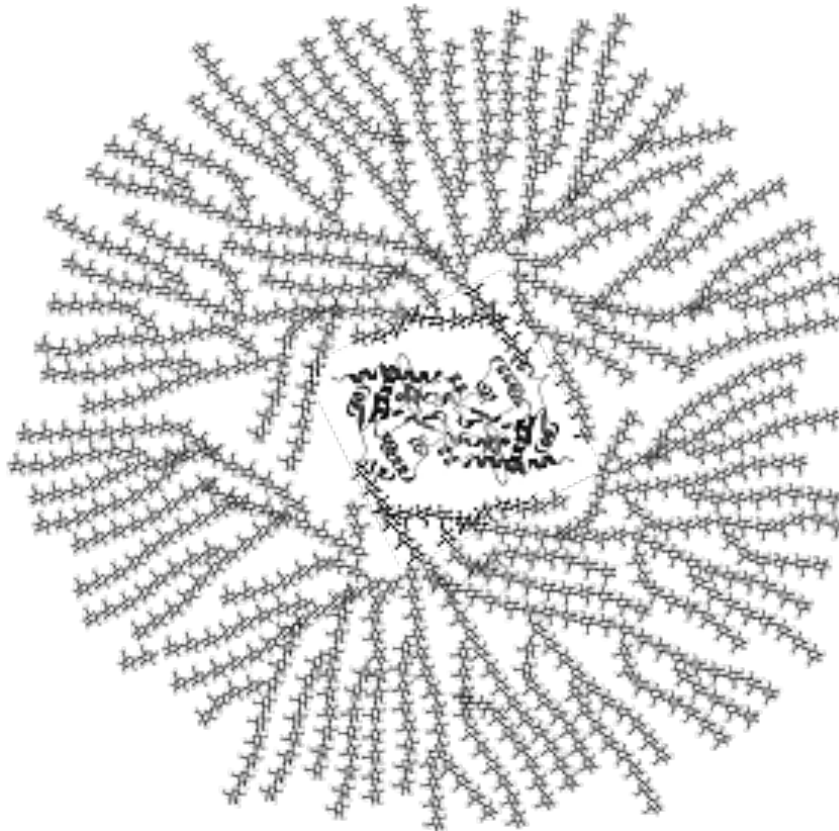


Fig. 2.1

(b) Explain how the structure of glycogen is related to its role in living organisms.

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[4]

(c) Describe how the structure of cellulose is different from the structure of glycogen.

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[3]

[Total: 11]

- 3 Catalase is an enzyme that catalyses the breakdown of hydrogen peroxide, a toxic waste product of metabolism.

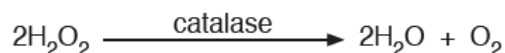


Fig. 3.1 shows the results of an investigation into the effect of hydrogen peroxide concentration on the rate of the catalase-controlled reaction, with and without the presence of two different inhibitors.

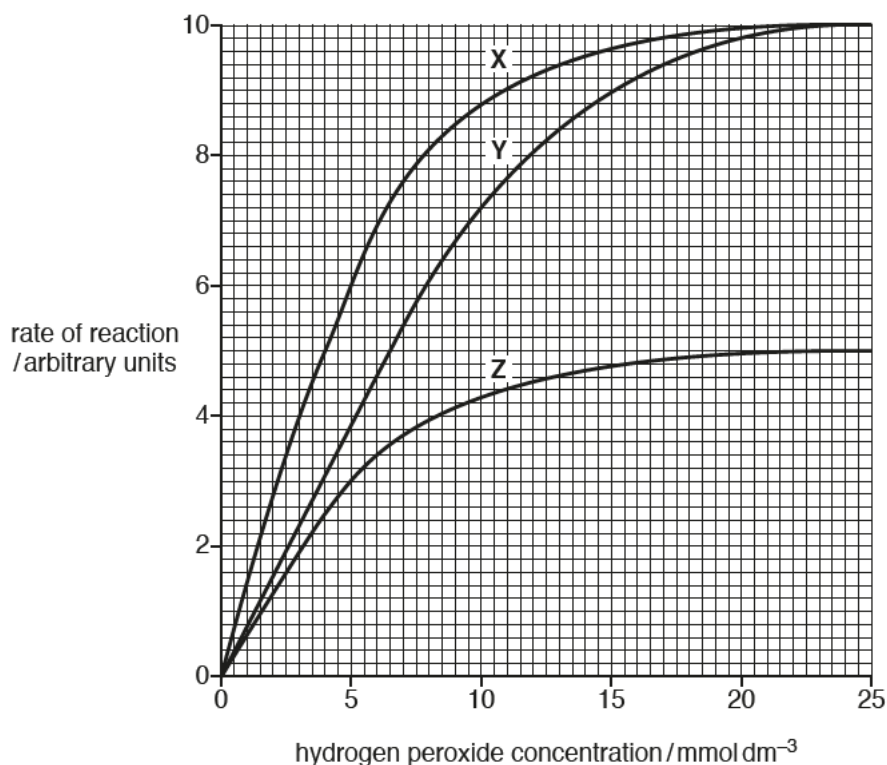


Fig. 3.1

- (a) The inhibitors used in the investigation have different modes of action.

Identify which of curves **X**, **Y** and **Z** are the results for:

- the reaction with the non-competitive inhibitor
- the reaction with the competitive inhibitor
- the reaction without any inhibitor.

non-competitive inhibitor

competitive inhibitor

without any inhibitor

- (b) With reference to Fig. 3.1, compare the maximum rate of reaction, $V_{\max}$, and the Michaelis-Menten constant, K_m , for curves **X**, **Y** and **Z**.

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[4]

- (c) Account for graph **Z**.

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[3]

[Total: 8]

- 4 Glucocorticoids (**S**), are a class of steroid hormones that bind to the glucocorticoid receptor (**GR**), and are crucial in the regulation of many genes. **S** binds to **GR**, activating **GR**. Activated **GR** binds to glucocorticoid response elements (**GREs**) within the promoter regions of target genes. This results in the recruitment of the chromatin remodelling complex, **BRG1** complex.

Fig. 4.1 shows the effect of **GR**-mediated gene expression and the effect of **BRG1** complex binding to the promoter region of the target gene.

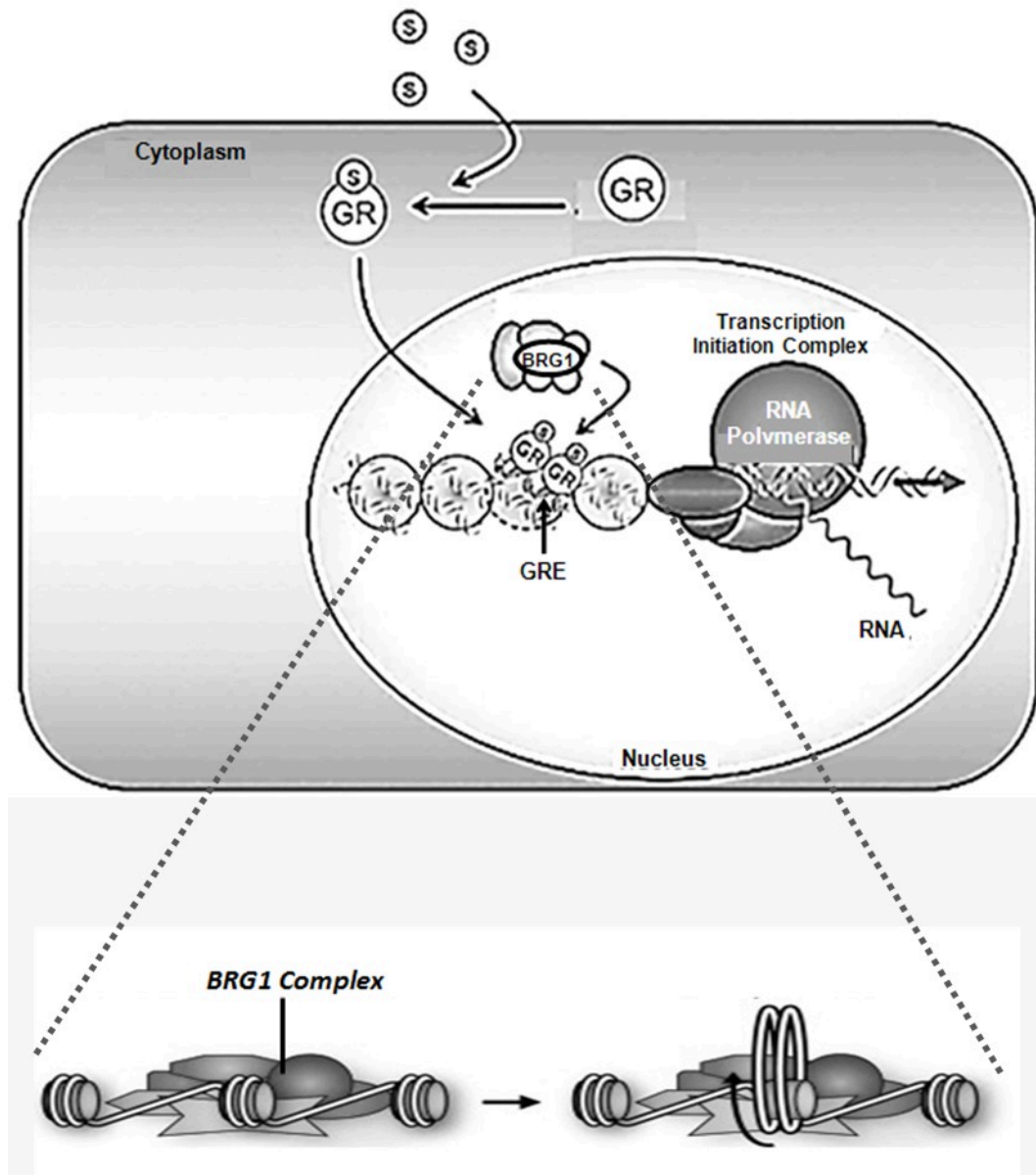


Fig. 4.1

(a) Explain how the steroid hormone is able to enter the cell.

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[1]

(b) **GRs** are known to have highly conserved regions which are structurally important for its function.

Describe two roles of **GRs**.

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[4]

(c) Levels of **GR** synthesis can be controlled translationally.

Describe how translational control can occur in eukaryotic gene expression.

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[3]

(d) State two differences between the cell structures used in translation in prokaryotes and eukaryotes.

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[2]

[Total: 10]

5 Fig. 5.1 shows the structure of the influenza virus.

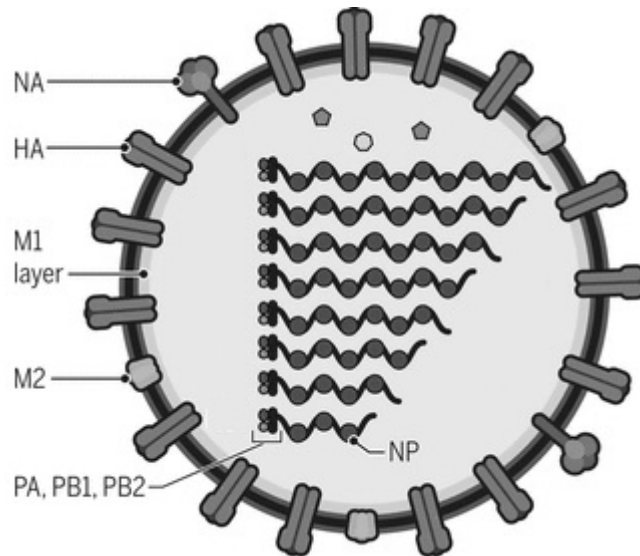


Fig. 5.1

(a) Identify the structures labelled:

label	name of structure
NA	
NP	

[1]

(b) The RNA segments containing genes PA, PB1 and PB2 shown in Fig. 5.1 code for subunit complexes that form RNA polymerases. Explain why such RNA segments are needed by the virus.

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[2]

- (c) The diagram below shows the structure of HA and positions of amino acid where changes are frequently observed in antigenic variants of the virus.

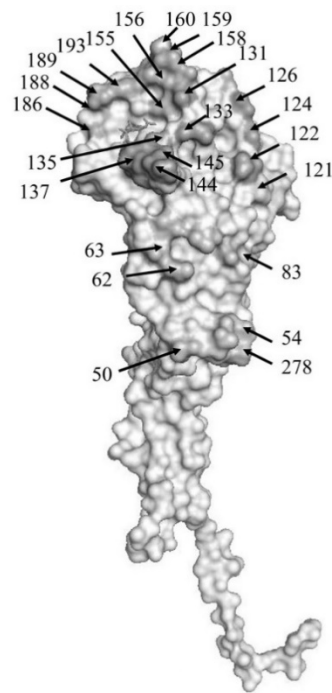


Fig. 5.2

- (i) Identify the type of antigenic changes that resulted in the variants.

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[1]

- (ii) Explain how the features of the virus contribute to these antigenic changes.

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[3]

(d) Contrast the reproductive cycle of influenza and HIV.

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[3]

[Total: 10]

- 6 Fig. 6.1 shows a stage in the mitotic cell cycle in an animal cell.

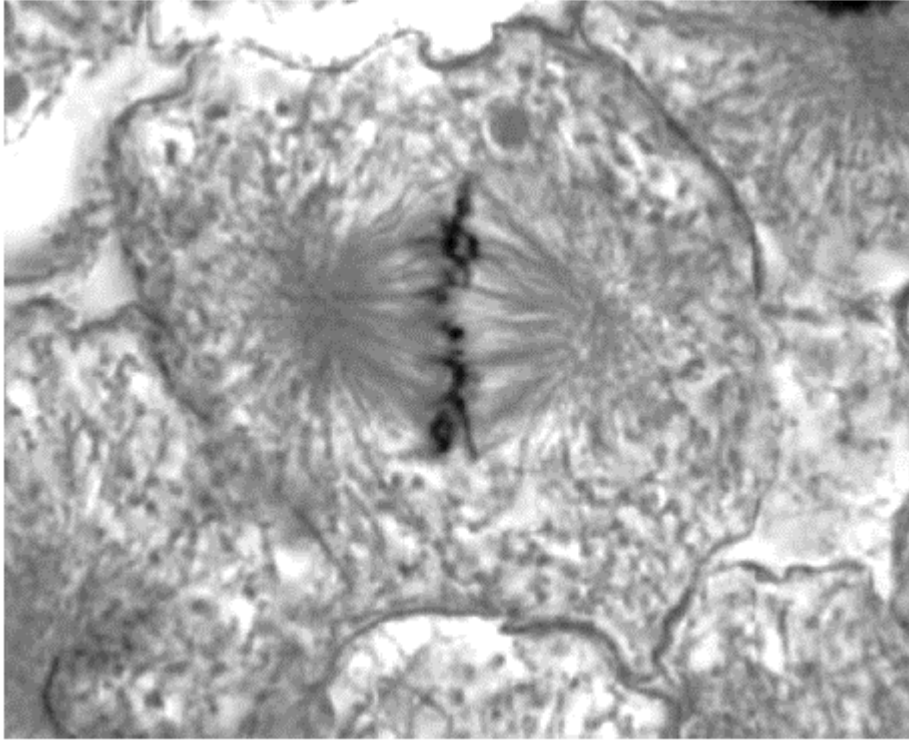


Fig. 6.1

- (a) With reference to Fig. 6.1,
(i) identify the stage of mitosis;

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[1]

- (ii) state two features which are characteristic of this stage.

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[2]

(b) Distinguish between the terms *haploid* and *diploid*.

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[2]

(c) Explain the importance of mitosis in organisms.

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[3]

(d) In many multicellular organisms, such as mammals, the time taken for the mitotic cell cycle varies considerably between different tissues but is very carefully controlled in each cell.

Suggest the importance of this control in mammals.

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[2]

[Total: 10]

- 7 A group of scientists discovered a new species of forest cockroach near their campus. Female cockroaches of this species are homogametic and the species follows the XO sex determination system.

Upon analysis of the samples they had collected, it was observed that there are three different eye colours in the species. These eye colours are determined by two genes, Gene B and Gene R, which are found on different chromosomes. One of the genes is present on the X chromosome while the other is found on an autosome. Fig 7.1 shows the formation of eye pigment in this forest cockroach species.

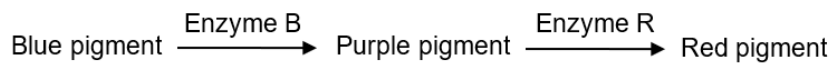


Fig. 7.1

The dominant B allele converts a blue pigment into a purple pigment while the dominant R allele converts the purple pigment into a red pigment.

- (a) (i) State the pattern of inheritance of the eye colour in the new species of forest cockroach.

[1]

- (ii) A true-breeding blue-eyed male was crossed to a true-breeding purple-eye female. All the F1 offspring from this cross had red eyes.

Deduce the gene that is located on the X chromosome and explain how you derived your answer.

[2]

- (iii) Draw a genetic diagram showing the expected ratio of phenotypes when the F1 progeny are crossed.

[4]

- (b) (i) Another species of the forest cockroach also has three eye colours but the expression of eye color in this species of forest cockroach is only controlled by one gene C found on an autosome.

It was found that a cross between true-breeding blue-eyed and red-eyed individuals produced all purple-eyed F1 progeny.

If the purple-eyed F1 progeny are mated with each other, deduce the expected F2 phenotypic ratio.

[1]

- (ii) 188 offspring were obtained in the F₂ generation and the phenotypes of the offspring were as follows:

40 cockroaches with blue eyes

100 cockroaches with purple eyes

48 cockroaches with red eyes

A χ^2 (chi-squared) test was performed to test the significance of the difference between the observed and expected results. The calculated $\chi^2 = 1.45$.

Part of the critical values of the chi-squared distribution is shown below.

Degrees of freedom	Probability		
	0.10	0.05	0.01
1	2.71	3.84	6.64
2	4.69	5.99	9.21
3	6.25	7.82	11.35
4	7.78	9.49	13.28

Explain whether your calculated χ^2 value in support the expected phenotypic ratio that you have obtained in (b)(i).

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[3]

[Total: 11]

8 (a) Photosynthesis is a complex process involving a light dependent stage and a light independent stage.

(i) Name the products of the light dependent stage that are needed in the light independent stage.

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[2]

(ii) Describe the role of chlorophyll b in photosynthesis.

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[2]

A student carried out an experiment to investigate the effect of light intensity and light wavelength on the rate of photosynthesis.

- An aquatic plant, *Elodea canadensis*, was put into a beaker containing sodium hydrogencarbonate solution as a source of carbon dioxide.
- To minimise changes in temperature, an LED lamp was used as a source of light.
- The lamp was switched on and the number of bubbles released by the aquatic plant in 1 minute was counted.
- The lamp was placed at seven different distances from the beaker to change light intensity.
- Five replicates were carried out at each lamp distance.
- All other variables were controlled.

- (b) The student calculated the light intensity for each distance (d) using $\frac{1}{d^2}$.
Table 8.1 shows the calculated light intensities for each distance.

Table 8.1

distance between plant and lamp / m	light intensity / $\frac{1}{d^2}$
0.025	1600
0.050	400
0.100	
0.150	44
0.200	25
0.250	16
0.300	11

Complete Table 8.1 by calculating the light intensity for distance 0.100m.

[1]

- (c) At each distance from the lamp, the experiment was repeated using a red filter in front of the lamp to give a different wavelength of light. The experiment was repeated using a blue filter and then using a green filter. Each filter transmitted the same light intensity.

The student calculated the mean rate of bubble production as a measure of the rate of photosynthesis.

Fig. 8.1 shows a graph of the results.

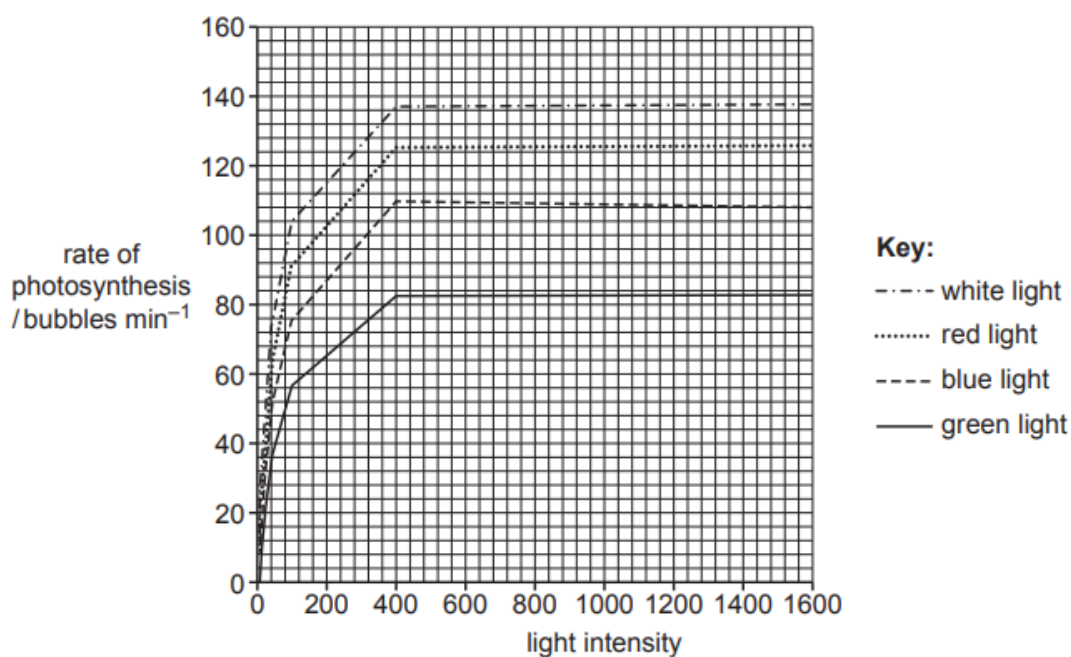


Fig. 8.1

(i) With reference to Fig. 8.1:

- state the range over which light intensity is the limiting factor
- explain why light intensity above this range is not limiting the rate of photosynthesis.

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[3]

(ii) At a light intensity of 1600, explain why different colour filters result in different rates of photosynthesis.

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[3]

[Total: 11]

- 9 (a) The European eel, *Anguilla anguilla*, is a fish. The sizes of eel populations tend to remain relatively stable despite eels producing large numbers of offspring.

Suggest **two** reasons why the population sizes of eels tend to remain relatively stable.

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[2]

- (b) Explain what is meant by the theory of evolution.

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[2]

- (c) The generation time of a species is the mean (average) time from one generation (parents) to the next generation (offspring). For example, the generation time of humans is about 25 years.

Fig. 9.1 shows a graph of the relationship between the rate of evolution and the generation time for a wide range of different species.

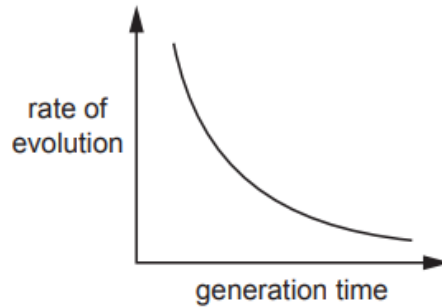


Fig. 9.1

Describe **and** explain the relationship shown in Fig. 9.1.

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[3]

- (d) The tuatara, *Sphenodon punctatus*, is a reptile that is native to New Zealand. It is found nowhere else in the wild.

Fig. 9.2 shows a tuatara.



Fig. 9.2

Tuataras have a slow growth rate and can live for over one hundred years. Fossil evidence shows that there has been little morphological change in the tuatara over the last 200 million years. This is a much lower rate of evolution than would be expected from the generation time of this species.

Suggest why the tuatara has remained largely unchanged over the last 200 million years.

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[3]

[Total: 10]

- 10** Dengue viruses (DENV) are responsible for millions of infections each year in tropical and subtropical areas of the world. According to the World Health Organization, dengue incidence has increased significantly over the past 50 years, turning this infection into the most important mosquito-borne disease in the world and a global health challenge. Fig. 10.1 shows the general global distribution of dengue fever in the year 2005.

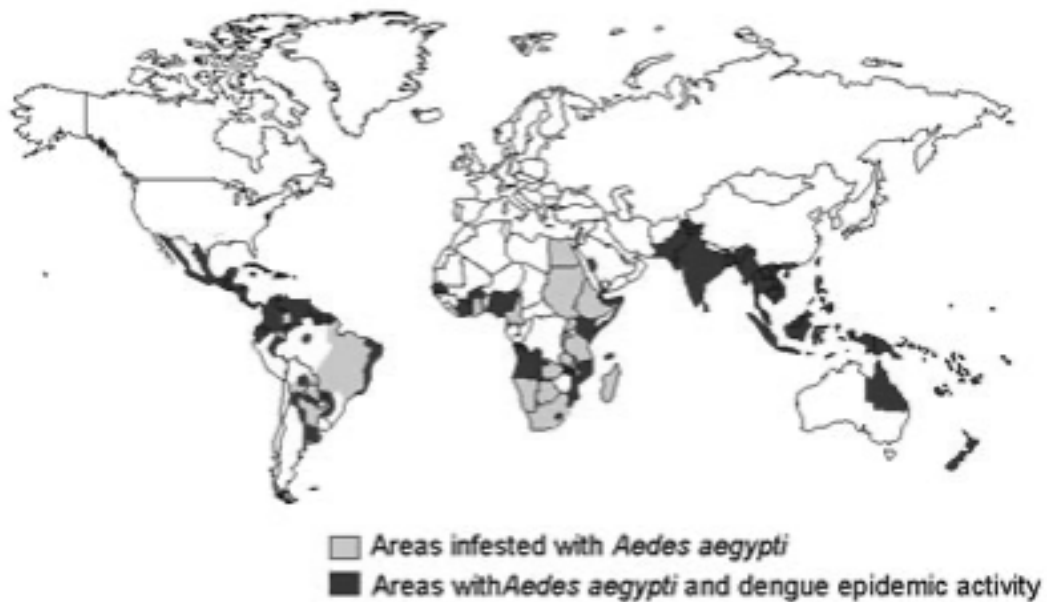


Fig. 10.1

- (a) (i)** Climate is one of the important factors that affects the distribution of dengue.

Predict and justify the expected distribution of dengue by the end of the 21st century.

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[2]

- (ii) Symptoms of dengue such as fever usually develop within 4 to 7 days after being bitten by an infected mosquito and is often associated with joint pain.

Explain how DENV may cause these symptoms.

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[2]

- (b) Outline the roles of T lymphocytes in the adaptive immune response after an infection with DENV.

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[4]

- (c) To some extent the range of the *Aedes* mosquito has also followed human expansion. Explain how this may be true.

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[2]

[Total: 10]